Where companies use Python, NumPy, Pandas, Matplotlib, and JupyterLab

Companies use these tools together to turn raw data into useful business decisions. Each tool has a different role, and they are often used by the same data analyst or data science team.

Example 1 — Manufacturing company

Problem: A factory machine unexpectedly stops working, interrupting production.

- Python: automate collection of machine sensor data.
- NumPy: calculate vibration and temperature statistics.
- Pandas: identify abnormal readings and compare machine performance.
- Matplotlib: plot vibration trends and temperature changes.
- JupyterLab: let engineers investigate faults and share analysis.

Outcome: Detect warning signs earlier and plan maintenance before likely failures.

Example 2 — Banking and financial services

Problem: A bank needs to identify suspicious transactions and monitor financial risk.

- Python: automate transaction analysis.
- NumPy: calculate transaction statistics and risk measures.
- Pandas: filter unusual transactions and group activity by account.
- Matplotlib: visualize transaction spikes and unusual patterns.
- JupyterLab: investigate cases and document findings for the risk team.

Outcome: Help investigators prioritize suspicious activity. Actual fraud detection also requires appropriate rules, domain expertise, and potentially machine-learning models.

Example 3 — Logistics and supply chain

Problem: Deliveries are frequently late, and vehicles travel more kilometres than necessary.

- Python: process delivery and GPS records.
- NumPy: calculate distances and delivery-time statistics.
- Pandas: compare delivery performance by route, driver, and day.

- Matplotlib: chart delays and identify problematic routes.
- JupyterLab: collaborate on delivery performance investigations.

Outcome: Identify delay patterns and support better route and dispatch planning. Advanced route optimization may also require mapping APIs and optimization algorithms.

Example 4 — Telecommunications

Problem: Customers experience slow internet speeds and dropped calls in particular areas.

- Python: collect network performance logs.
- NumPy: calculate packet-loss and latency statistics.
- Pandas: compare network performance across towers and time periods.
- Matplotlib: visualize outages, latency spikes, and network trends.
- JupyterLab: help network engineers investigate recurring incidents.

Outcome: Identify affected network segments and support capacity planning and troubleshooting.

Example 5 — E-commerce inventory management

Problem: Popular products run out of stock while other products remain unsold in warehouses.

- Python: automate daily inventory reports.
- NumPy: calculate demand averages and variability.
- Pandas: combine stock, orders, returns, and supplier records.
- Matplotlib: plot demand trends and stock levels.
- JupyterLab: analyze stockout patterns and share replenishment recommendations.

Outcome: Improve inventory planning and reduce lost sales and excess stock.